

HW 8 – The Geometric Distribution

1) For the following situations, decide if it is a binomial setting, a geometric setting, or neither. Explain your answer.

a) You keep drawing cards out a deck, without replacement, until an ace is drawn.

b) You roll a dice 20 times and record the number of sixes you have rolled.

c) Crest claims that 40% of Americans use their toothpaste. You take a random sample of 50 Americans and count how many use Crest toothpaste.

d) You flip a coin until you get tails.

e) 5% of the tomatoes at a farmer's market have imperfections on them. You randomly choose one tomato at a time until you find one with an imperfection.

f) 5% of the tomatoes at the farmer's market have imperfections on them. You randomly choose 20 tomatoes and count the number of imperfections on them.

2) In a recent Pew study, it was found that 28% of all homes have at least one dog. Let X = the number of houses you look at before finding a house with a dog. Find and interpret the following probabilities.

a) $P(X = 3)$

b) $P(X < 5)$

c) $P(X > 4)$

d) $E(X)$

3) Zach's teammate, Taylor, is the quarterback for the football team. She can complete 65% of her passes.

For the following table, fill in the missing pieces.

<i>Situation</i>	<i>Probability Notation</i>	<i>Formula</i>	<i>Value</i>
What is the probability that it takes Taylor 3 incomplete passes before she has a completion?			
	$P(X < 3)$		
		$(0.35)^4(0.65)$	
		$1 - (P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5))$	